

Vocabulary

Domain restriction: an x -value that must be excluded so the function stays defined.

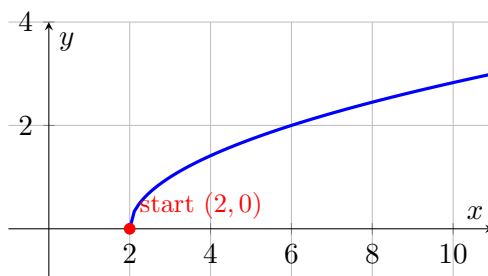
Radicand: the expression underneath a radical (root) symbol; for a square root it must be ≥ 0 .

We do together. Find the domain and range of $f(x) = \sqrt{x - 2}$.

Step 1. The radicand is $x - 2$; a square root needs a non-negative radicand, so set up $x - 2 \geq 0$.

Step 2. Solving gives the domain: $x \geq 2$.

Step 3. At the smallest allowed input $x = 2$, $f(2) = \underline{0}$, and the graph only rises after that, so the range is $y \geq 0$.



$y = \sqrt{x - 2}$ — domain $x \geq 2$, range $y \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Find the domain of $f(x) = \frac{1}{x + 6}$.

$x \neq -6$ (all reals except -6)

b) Find the domain of $f(x) = \sqrt{10 - x}$.

$x \leq 10$

Summary (fill in): Two traps restrict the domain: a denominator can never equal 0, and a radicand (under an even root) can never be negative.